

2016.0 RANGE ROVER (LG), 205-02

REAR DRIVE AXLE/DIFFERENTIAL

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For additional information, refer to the relevant description and operation section of the workshop manual.

REFER to: [Rear Drive Axle and Differential](#) (205-02 Rear Drive Axle/Differential, Description and Operation).

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is NOT acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle

1. Verify the customer concern
2. Visually inspect for obvious signs of mechanical or electrical damage

Visual Inspection

MECHANICAL

ELECTRICAL

▪ Differential oil leakage ▪ Sensor installation	▪ Power supply ▪ Fuse(s) ▪ Wiring harness physical damage or water ingress ▪ Loose or corroded electrical connectors ▪ CAN circuits ▪ Sensors ▪ Rear differential control module
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3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step
4. If the cause is not visually evident, verify the symptom and refer to the symptom chart, alternatively check for diagnostic trouble codes (DTCs) and refer to the DTC index

SYMPTOM CHART

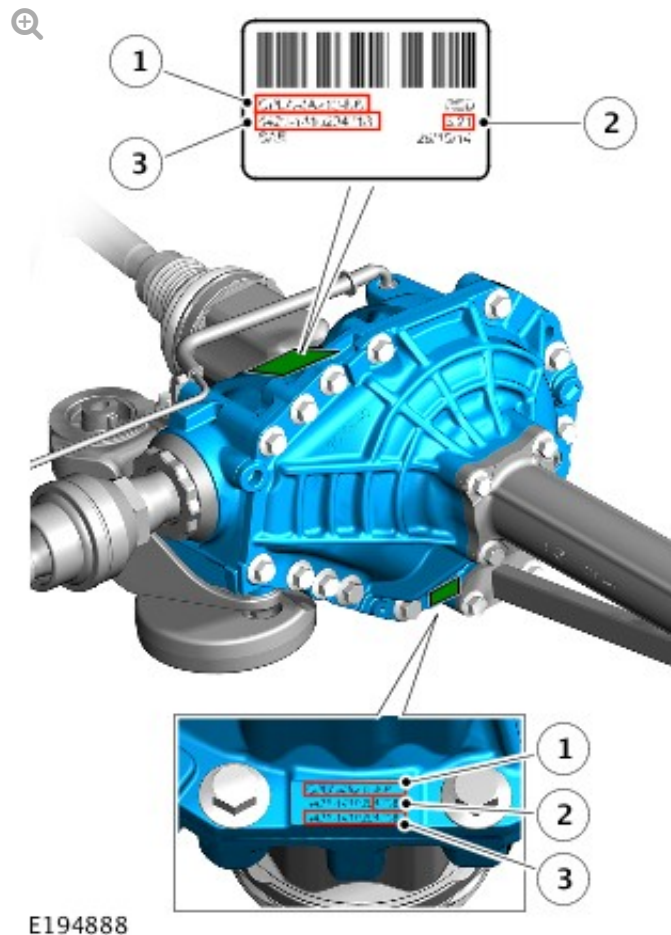
SYMPTOM	POSSIBLE MESSAGE	POSSIBLE CAUSES	ACTION
Rumbling noise from the rear of the vehicle varying at different vehicle speed and load	▪ N/A	▪ Rear differential internal failure ▪ Road noise ▪ Worn or damaged driveshaft joint ▪ Wheel bearing ▪ Another component contacting the front/rear drive halfshaft	Using the manufacturer approved diagnostic system run application Noise, vibration and harshness diagnostic test - Rear differential

Running at reduced capability with fault present	▪ N/A	▪ Water ingress to wiring harness or connectors ▪ Reduced differential capability ▪ Differential increased tolerances ▪ Internal fault ▪ Rear Differential Control Module fault (Rear Differential Control Module)	Check for stored DTCs and refer to the DTC index. Visually inspect the wiring harness and connectors for water ingress. Refer to the warranty policy and procedures manual if a module is suspect.
Rear differential overheat	▪ Rear Axle Overheat	▪ Oil level incorrect ▪ Oil level incorrect specification ▪ Sensor fault ▪ Internal fault	Check for stored DTCs and refer to the DTC index. Check for correct oil quantity and specifications. Refer to the relevant section of the workshop manual.
Excessive steering effort	▪ Rear Axle Fault	▪ Sticking or stuck actuator	Check for stored fault codes and refer to DTC index.

TCM DTCS P0721-62 OR P0721-64 - DIFFERENTIAL/FINAL DRIVE RATIO CHECKS

In the case of customer reports of vibration that may be traced to either the transmission or the transfer case **and/or** a "Gearbox Fault" message

displayed on the instrument cluster when driving at speeds in excess of 120 kph **and/or** the flagging of either DTC P0721-62 or DTC P0721-64 in the Transmission Control Module, then this may indicate that the vehicle has been fitted with an incorrect front and/or rear differential. In such cases, there may be a mismatch in the final drive ratio between front and rear.



1 - Land Rover Part Number

2 - Ratio

3 - Serial Number

Range Rover/Range Rover Sport (L405/L494) Front & Rear Differential Parts Matrix

To check the correct differential units are fitted, cross check the part numbers found on the labels attached to the front and rear differential casings (see illustration above) with the part numbers listed in the tables below:

VEHICLE	ENGINE	RATIO	PART DESCRIPTION	LAND ROVER PART NUMBER
L494	GTDi 2.0L Petrol	4.44	Front	FPLA 3017 BB
			Rear - Open	FPLA 4A213AB
	V6 S/C 3.0L Petrol	3.73	Front	DPLA 3017 EE
			Rear - Open	DPLA 4A213JB
	V8 5.0L S/C Petrol	3.31	Front	CPLA 3017 CF
			Rear - Electronic	DPLA 4A213GB

VEHICLE	ENGINE	RATIO	PART DESCRIPTION	LAND ROVER PART NUMBER
L494	TDV6 3.0L Diesel	3.21	Front	CPLA 3017 BF
			Rear - Open	CPLA 4A213BC
			Rear - Electronic	DPLA 4A213FC
	TDV6 3.0L Diesel HEV	3.21	Front	EPLA 3017 AD
			Rear - Electronic	EPLA 4A213FB
	TDV8 4.4L Diesel	2.73	Front	CPLA 3017 AF
			Rear - Electronic	DPLA 4A213 EC

VEHICLE	ENGINE	RATIO	PART DESCRIPTION	LAND ROVER PART NUMBER
L405	V6 S/C 3.0L Petrol	3.73	Front	DPLA 3017 EE
			Rear - Open	DPLA 4A213JB
	V8 5.0L Petrol	3.55	Front	CPLA 3017 DF
			Rear - Open	CPLA 4A213DB
	V8 5.0L S/C Petrol	3.31	Front	CPLA 3017 CF
			Rear - Open	CPLA 4A213CB
			Rear - Electronic (optional)	DPLA 4A213GB

VEHICLE	ENGINE	RATIO	PART DESCRIPTION	LAND ROVER PART NUMBER
L405	TDV6 3.0L Diesel HEV	3.21	Front	EPLA 3017 AD
			Rear - Open	EPLA 4A213BB
			Rear - Electronic (optional)	EPLA 4A213FB
	TDV8 4.4L Diesel	2.73	Front	CPLA 3017 AF
			Rear - Open	CPLA 4A213AC
			Rear - Electronic (optional)	DPLA 4A213 EC

If the part numbers of the front and/or rear differential units match those in the tables above, then follow the diagnostic steps outlined in the relevant DTC Index and elsewhere in TOPIx (if further technical support/advice is required, an additional TA should be raised)

If the part numbers of the front and/or rear differential units do not match those in the tables above, follow the steps outlined below:

NOTE:

Submit an EPQR (Electronic Product Quality Report) whenever an incorrect differential has been reported/discovered

1. Raise a TA/FRED (Technical Assistance Request/Fixed Right Enhanced Diagnostics) for pre-approval to replace the differential
2. Subject to TA/FRED approval, replace the incorrect axle(s). Clear all DTCs and perform a road test (ensure that the vehicle is driven at speeds in excess of 120 kph in order to trigger any potential re-occurrence of TCM DTCs)
3. If no faults or symptoms are present, return the vehicle to the customer

4. If vibration is present or if DTCs P0721-62 or P0721-64 are flagged in the Transmission Control Module, then follow the diagnostic steps outlined in TOPIx (if further technical support/advice is required, an additional TA should be raised)

D T C I N D E X

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00. REFER to: [Diagnostic Trouble Code Index - DTC: Rear Differential Control Module](#) (100-00 General Information, Description and Operation).